

Scalability & Future Plan

Date	03 July 2026
Team ID	SWTID-2026-4593
Project Name	Machine Learning–Based Credit Card Approval Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team’s roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited historical data range (e.g., only 2 years).	Reduced model training dataset diversity and potential for temporal bias.	Medium
2	High prediction latency during peak usage hours (single worker).	Degraded user experience and increased request timeouts on the Flask app.	High
3	Basic feature engineering (missing income-to-debt interaction features).	Sub-optimal model accuracy for certain demographic groups and edge cases.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance, synchronous Flask application.	Implement Gunicorn with multi-workers and autoscaling using Docker/Render horizontally scaled workers.
2	Data Storage	Single, non-replicated local MySQL instance.	Migrate to a managed MySQL cluster (e.g., on Render) with read-replicas for load balancing and high availability.
3	Performance	Model inference executes on the main application thread.	Offload inference requests to a separate, asynchronous task queue (e.g., Celery/Redis) and optimize model with ONNX.
4	Security	Basic API key authentication with unencrypted PII storage.	Implement OAuth 2.0 with JWT, role-based access control, full database-level encryption for all sensitive user data.

Future Roadmap:

Phase	Planned Feature	Target Timeline	Expected Impact
Phase 1 (Short-term: ~6 mos)	Automated Model Retraining & Hyperparameter Tuning Pipeline	03 Jan 2027	Maintained model performance over time and automated continuous optimization.
Phase 2 (Mid-term: ~12 mos)	Feature Enrichment (interaction and demographic features)	03 July 2027	Improved model accuracy by 5-8% on edge case predictions.
Phase 3 (Long-term: ~18 mos)	Explainable AI (XAI) Dashboard for administrators	03 Jan 2028	Increased transparency in prediction decisions and enabling fairness audibility.

Scalability & Future Plan

Scalability Area / Phase	Proposed Upgrade / Enhancement	Expected Performance Impact	Target Timeline
Phase 2	Migrating deployed model services to auto-scaling cloud infrastructure (e.g., ECS with Fargate) and database replication (e.g., MySQL read-replicas). Implement content delivery network (CDN) for static assets.	Improved system responsiveness during peak usage periods. Higher concurrent user support. Reduced application latency. Enhanced data availability and fault tolerance.	Q1 2027
Phase 3	Integrate continuous model monitoring and drift detection APIs. Explore integration with advanced anomaly detection models or third-party datasets for enhanced prediction features.	Proactive identification of model performance issues. Potentially higher prediction accuracy. Expanded system capabilities.	Q3 2027
Phase 4	Hardening the system security layer. Developing a secured mobile application interface. Expanding model applicability to new related product domains. Implement CI/CD for automated pipeline.	Reduced vulnerability to cyber-attacks. Increased market reach and accessibility. Higher overall system reliability. Enhanced business value and strategic alignment.	Q1 2028